

The limit of vector-valued smoothing for Sidon sets is $2\sqrt{2}/3$

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Abstract

Let $h(N)$ be the maximum size of a Sidon set contained in $\{1, \dots, N\}$. The best upper bounds of the form $h(N) \leq N^{1/2} + \gamma \cdot N^{1/4} + O(1)$ now come from numerical certificates in the vector-valued smoothing framework of Hou and Zhao, which reduced γ to 0.9435. We determine the limit of this method. For the framework and its natural two-sided extension, which admits asymmetric kernels, every certificate has $\gamma \geq 2\sqrt{2}/3 = 0.94280904\dots$, and in the two-sided framework this is sharp: the linear kernel $\rho(x) = 2x$, with boundary weights built from the renewal measure of the uniform distribution, attains $2\sqrt{2}/3$ in the continuum limit, and its discretizations approach it at rate $O(\log m / m^2)$. The lower bound comes from a single dual multiplier, $2w$ times the renewal measure of the uniform distribution on $[0, w]$ with w chosen from the kernels' energy. This multiplier places a constant weight on the kernels' autocorrelation, so the bound it certifies depends on the kernels only through their L^2 norm; the constancy reduces to the identity $\int_0^\infty (r(x) - 2)^2 dx = 1/3$ for the renewal density r , which we prove. A certificate verified in exact rational arithmetic gives $h(N) \leq N^{1/2} + 0.9428096 \cdot N^{1/4} + O(1)$, within 5×10^{-7} of the limit. Arguments of this type therefore cannot improve the constant beyond $2\sqrt{2}/3$, nor the exponent $1/4$ in Erdős Problem 30. We also show that the natural semidefinite relaxation that keeps the variance term discarded by the method, at the level of pair statistics, has the same second-order constant: every certificate bounds it, and explicit feasible points come within about 1 of $N^{1/2} + (2\sqrt{2}/3) \cdot N^{1/4}$ for N up to 10^7 .

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1. Introduction

A set of integers is a *Sidon set* if all of its pairwise sums $a + b$ with $a \leq b$ are distinct, equivalently if its non-zero differences are distinct. Let $h(N)$ be the largest size of a Sidon set contained in $\{1, \dots, N\}$. Erdős and Turán [7] proved $h(N) \leq N^{1/2} + O(N^{1/4})$, and Lindström [12] obtained $h(N) \leq N^{1/2} + N^{1/4} + 1$. Erdős Problem 30 [2, 13] asks whether $h(N) = N^{1/2} + O_\varepsilon(N^\varepsilon)$ for every $\varepsilon > 0$. In eighty-five years the exponent $1/4$ has not moved; the constant in front of it has.

upper bound $h(N) \leq N^{1/2} + \gamma \cdot N^{1/4} + O(1)$	γ	year
Erdős–Turán; Lindström	1	1941, 1969
Balogh–Füredi–Roy [1]	≈ 0.998	2021
O’Bryant [14]	0.99703	2022
Carter–Hunter–O’Bryant [4]	0.98183	2023
Carter, Georgiev, Gómez-Serrano, Hunter, O’Bryant, Tao, Wagner (announced)	0.97633	2026
Hou–Zhao [10], vector-valued smoothing	0.9435	2026
this paper: certificate of Theorem 2	0.9428096	2026
this paper: limit of the method (Theorem 1)	$2\sqrt{2}/3 = 0.9428090\dots$	—

Table 1. The second-order constant. Years are those of first public appearance.

The most recent improvements are of a different character from their predecessors. Hou and Zhao [10] recast the whole family of smoothing arguments as a finite optimization problem: a bound is certified by kernels, boundary weights and mixing weights satisfying an explicit system of linear inequalities, and the constant is $\gamma = \sqrt{ab}$ for two quadratic quantities a and b (Proposition 2.1). Improvements are then found by optimizing. This paper determines exactly how far that road goes. Write γ^*_{sym} for the infimum of γ over the certificates of [10], whose kernels are symmetric, and γ^* for the infimum over a two-sided extension that admits asymmetric kernels (Proposition 2.3), so that $\gamma^* \leq \gamma^*_{\text{sym}}$.

Theorem 1. (a) Every certificate, symmetric or two-sided, satisfies $\gamma \geq 2\sqrt{2}/3$; thus $\gamma^*_{\text{sym}} \geq \gamma^* \geq 2\sqrt{2}/3 = 0.94280904\dots$. (b) For two-sided certificates the bound is sharp: $\gamma^* = 2\sqrt{2}/3$.

Theorem 2 (computational). *The linear kernel $p(x) = 2x$, discretized at resolution $m = 512$ with boundary depth $L = 8$ and exactly optimal boundary weights, gives a two-sided certificate whose 8194 covering inequalities and whose constants a, b, γ^2 have been verified in exact rational arithmetic, with $\gamma < 0.942809522$. Consequently, by Proposition 2.3, $h(N) \leq N^{1/2} + 0.9428096 \cdot N^{1/4} + O(1)$.*

Arguments of this type can therefore never certify a constant below $2\sqrt{2}/3$, and in particular can never reach an error term of size $o(N^{1/4})$; the remaining room below the constant of Theorem 2 is less than 5×10^{-7} .

The idea of the proof. For fixed kernels the optimization over boundary weights is a convex quadratic program, and any nonnegative multiplier μ on its constraints gives a lower bound on b by weak duality (Lemma 3.1). The multiplier that proves Theorem 1 is

$$\mu_w = 2w \cdot \sum_{n \geq 0} \{n \geq 0\} U_w^{*n}, \quad (1.1)$$

twice w times the renewal measure of the uniform distribution U_w on $[0, w]$: an atom of mass $2w$ at the origin, density $2e^{u/w}$ on $[0, w)$, and so on, settling to the flat value 4. Its key property (Lemma 4.3) is that the weight it places on the autocorrelation R of the kernels is constant — equal to $8w/3$ — at every lag shorter than w . The bound on b therefore depends on the kernels only through $a = R(0)$:

$$b \geq 4w/3 - (w^2/2) \cdot a \text{ for every } w > 0, \quad (1.2)$$

and $w = 4/(3a)$ gives $b \geq 8/(9a)$, that is $a \cdot b \geq 8/9$. The constancy of the weight reduces to one identity for the renewal density r of the uniform distribution, $\int_0^\infty (r(x) - 2)^2 dx = 1/3$, which we prove from the renewal equation (Lemma 4.2). Sharpness comes from the linear kernel $2x$: with boundary weights built from μ , every covering constraint is active and the dual bound is attained (Theorem 5.1).

How the result was found. The paper grew out of numerical experiments reported in Section 6. Resolution scans located the floor of the framework near 0.9428091; the optimal dual multiplier read off the quadratic programs turned out to be (1.1) with $w = 1$; feeding that multiplier back into the optimality conditions identified the linear kernel; and a scale-covariant family of multipliers fitted per certificate never certified less than $8/9$ on any kernel tried, which pointed to the identity behind (1.2).

Dependence on [10]. The implication from a certificate to a bound on $h(N)$ is Theorem 2.1 of Hou and Zhao [10] (Proposition 2.1 below), and its two-sided extension is Proposition 2.3. We prove the latter in full in Appendix C, following their argument, so the statements about $h(N)$ in this paper rest on a proof given here, while [10] remains the source of the framework. The proof in Appendix C also makes the $O(1)$ term explicit. We compared it

with the proof in [10] step by step, and checked every step by direct computation on explicit Sidon sets. Theorem 1 is a statement about the optimization problem and holds regardless. We do not know whether symmetric certificates reach $2\sqrt{2}/3$: the extremal kernel is asymmetric, the best symmetric certificate we have gives $\gamma < 0.942810893$, and our numerics at boundary depth 4 put the symmetric floor within 10^{-7} of $2\sqrt{2}/3$ but cannot decide equality (Section 6.4).

Section 2 sets up the framework, its two-sided extension and its continuum relaxation. Section 3 derives the dual bound. Section 4 proves the renewal identities and the lower bound. Section 5 proves sharpness. Section 6 reports the numerical picture. Section 7 studies the natural semidefinite relaxation that keeps the variance term the method discards, and shows that it has the same second-order constant. Section 8 discusses consequences.

2. The certificate framework

2.1 Discrete certificates

Fix integers $m \geq 1$ (bins per kernel), $L \geq 1$ (boundary blocks) and $K \geq 1$ (kernels). A *certificate* at resolution (m, L, K) consists of symmetric nonnegative kernels $p^{(k)} \in \mathbb{R}^m$ with $\sum_i p^{(k)}_i = 1$, boundary weights $w^{(k)} \in \mathbb{R}^{(Lm)}$ extended by $w^{(k)}_j = 1$ for $j \geq Lm$, and mixing weights $\lambda_k \geq 0$ with $\sum_k \lambda_k = 1$, subject to

$$\sum_k \lambda_k \sum_{i=0}^{m-1} p^{(k)}_i w^{(k)}_{\{q+i\}} \geq 1 \quad (q = 0, 1, \dots, Lm). \quad (2.1)$$

Its constants are

$$a = m \cdot \sum_k \lambda_k \sum_i (p^{(k)}_i)^2, \quad b = 1 + 2 \left(\sum_k \lambda_k \sum_{j < Lm} (w^{(k)}_j)^2 / m - L \right). \quad (2.2)$$

Proposition 2.1 (Hou-Zhao [10, Theorem 2.1]; see also Appendix C). *If a certificate satisfies (2.1) and $b > 0$, then $h(N) \leq N^{1/2} + \sqrt{ab} \cdot N^{1/4} + O(1)$.*

Definition 2.2. $\gamma^*_{\text{sym}} := \inf \sqrt{ab}$ over all certificates at all resolutions.

2.2 Two-sided certificates

A *two-sided certificate* has kernels $p^{(k)}$ that are nonnegative with $\sum_i p^{(k)}_i = 1$ but need not be symmetric, mixing weights as before, and for each k two boundary vectors $u^{(k)}, v^{(k)} \in \mathbb{R}^{(Lm)}$, extended by 1, subject to

$$\begin{aligned} \sum_k \lambda_k \sum_i p^{(k)}_i u^{(k)}_{\{q+i\}} &\geq 1 \quad (q = 0, \dots, Lm), \\ \sum_k \lambda_k \sum_i p^{(k)}_{\{m-1-i\}} v^{(k)}_{\{q+i\}} &\geq 1 \quad (q = 0, \dots, Lm), \end{aligned} \quad (2.3)$$

with a as in (2.2) and $b = 1 + (1/m) \sum_k \lambda_k \sum_{\{j < Lm\}} [(u^{(k)}_j)^2 + (v^{(k)}_j)^2 - 2]$. A symmetric certificate with $u = v = w$ is a two-sided certificate with the same a and b .

Proposition 2.3. *If a two-sided certificate satisfies (2.3) and $b > 0$, then $h(N) \leq N^{1/2} + \sqrt{ab} \cdot N^{1/4} + O(1)$.*

Proof. A complete proof is given in Appendix C. It follows the proof of [10, Theorem 2.1] step by step, and symmetry of the kernels is used there in one place only: the right-boundary case of the covering claim [10, (6)], where the identity $K_s = K_{H-1-s}$ reduces that case to the left one. Using $v^{(r)}$ on the right boundary region, and computing that case with the reversed kernel, yields the second system in (2.3) instead. The energy bound [10, (5)] needs only that the kernels are nonnegative, and the weight identity [10, (10)] becomes $N - 1 + bH$ with the two-sided b . The positivity argument of [10, Section 3.2], which also uses symmetry, is not needed: $b > 0$ holds by Theorem 4.4. ■

Definition 2.4. $\gamma^* := \inf \sqrt{ab}$ over all two-sided certificates at all resolutions. Thus $\gamma^* \leq \gamma^*_{\text{sym}}$.

2.3 The continuum relaxation

A *continuum certificate* consists of probability densities ρ_k on $[0,1]$ (not necessarily symmetric), mixing weights λ_k , and for each k two measurable profiles $\omega_k, \omega'_k : [0,\infty) \rightarrow \mathbb{R}$ with $\omega_k - 1$ and $\omega'_k - 1$ in $L^1 \cap L^2$, such that for every $u \geq 0$

$$\sum_k \lambda_k \int_0^1 \rho_k(x) \omega_k(u+x) dx \geq 1, \sum_k \lambda_k \int_0^1 \bar{\rho}_k(x) \omega'_k(u+x) dx \geq 1, \quad (2.4)$$

where $\bar{\rho}(x) = \rho(1-x)$. Its constants are $a = \sum_k \lambda_k \int_0^1 \rho_k^2$ and $b = 1 + \sum_k \lambda_k \int_0^\infty (\omega_k^2 + \omega_k'^2 - 2)$.

Lemma 2.5. *Every discrete certificate, symmetric or two-sided, is a continuum certificate with the same a and b .*

Proof. Read kernels and boundary vectors as step functions on cells of width $1/m$. The constants coincide with (2.2). At a shift $u = (q+s)/m$ with $0 \leq s < 1$ the left-hand sides of (2.4) are $(1-s)C_q + sC_{q+1}$, where C_q is the left-hand side of the q -th discrete inequality (with $C_q = 1$ for $q > Lm$), and so are at least 1. ■

Hence any lower bound for continuum certificates bounds every discrete certificate at every resolution, with no limiting argument.

3. A dual bound

For a continuum certificate write $R = \sum_k \lambda_k \rho_k \star \tilde{\rho}_k$ for the mixed autocorrelation, $\tilde{\rho}(x) = \rho(-x)$. R is even, nonnegative, supported in $[-1,1]$, $\int R = 1$ and $R(0) = a$. Put

$$G = \int |s| R(s) ds, T(v) = \int_{-v}^{\infty} R(s) ds \quad (v \geq 0). \quad (3.1)$$

A *multiplier* is a nonnegative measure on $[0, \infty)$ of the form $\mu = 4 \cdot \text{Leb} + \nu$, with ν a finite signed measure.

Lemma 3.1. *For every continuum certificate and every multiplier $\mu = 4 \cdot \text{Leb} + \nu$,*

$$b \geq G + \int T dv - \frac{1}{8} \iint R(u - v) \nu(du) \nu(dv). \quad (3.2)$$

Equality holds if both inequalities in (2.4) are equalities at every u in the support of μ and $\omega_k = \frac{1}{4}(\mu \star \rho_k)$, $\omega'_k = \frac{1}{4}(\mu \star \tilde{\rho}_k)$ for every k , where $(\mu \star \rho)(t) = \int \rho(t - u) \mu(du)$.

Proof. Write $e_k = \omega_k - 1$, so that the first end contributes $\sum_k \lambda_k \int (2e_k + e_k^2)$ to $b - 1$, and its covering condition reads $\sum_k \lambda_k \int \rho_k(x) e_k(u + x) dx \geq 0$. Integrate this condition against $\frac{1}{2}\mu$, which is nonnegative, and subtract; by Fubini the result is

$$\sum_k \lambda_k \int_0^{\infty} [2e_k + e_k^2 - \frac{1}{2} e_k(t) \cdot (\mu \star \rho_k)(t)] dt \geq - \sum_k \lambda_k \int_0^{\infty} (\frac{1}{4}(\mu \star \rho_k)(t) - 1)^2 dt,$$

the inequality being pointwise minimization in $e_k(t)$, attained at $e_k = \frac{1}{4}(\mu \star \rho_k) - 1$.

Since $4 \cdot \text{Leb} \star \rho_k = 4C_k$, with C_k the distribution function of ρ_k , the integrand is $(\frac{1}{4}(\mu \star \rho_k) - (1 - C_k))^2$. Expanding, $\int (\mu \star \rho_k)^2 = \iint r_k(u - v) \nu(du) \nu(dv)$, with $r_k = \rho_k \star \tilde{\rho}_k$; $\int (\mu \star \rho_k)(1 - C_k) = \int P(Y_k - X_k > u) \nu(du)$, with X_k, Y_k independent with density ρ_k ; and $\int (1 - C_k)^2$ is a constant. At the second end ρ_k is replaced by $\tilde{\rho}_k$. This changes neither r_k nor $P(Y_k - X_k > u)$, since X_k and Y_k are exchangeable, and turns the constant into $\int C_k^2$. The two constants sum to $1 - 2 \int C_k(1 - C_k) = 1 - E|X_k - Y_k|$. Summing over ends and kernels gives (3.2); the equality conditions are those of the two inequalities used. ■

Two consequences are worth recording. First, the right-hand side of (3.2) depends on the kernels only through R . In the discrete boundary program, where strong duality holds, the same is true of the optimal b itself (the end costs depend on the kernels' means, and these cancel between the two ends); reversing any kernel of a two-sided certificate changes b by less than 10^{-13} in our computations. Asymmetric kernels therefore matter only through the autocorrelations they can produce. Second, the one-atom multiplier $\mu = 4 \cdot \text{Leb} + \beta \delta_0$ with $\beta = 2/a$ gives $b \geq G + 1/(2a)$. Combined with the sharp inequality $\|\rho\|_2^2 \cdot E|X - Y| \geq \pi^2/32$, whose extremals are cosine arches (a short proof: write both factors through

the quantile function of p and apply Cauchy–Schwarz), this yields $a \cdot b \geq (\pi^2 + 16)/32$, i.e. $\gamma \geq 0.89912$. The rest of the paper replaces the atom by the full renewal measure (1.1), which closes the gap.

4. The renewal multiplier and the lower bound

4.1 The renewal density of the uniform distribution

Let U be the uniform density on $[0,1)$ and $M = \sum_{n \geq 0} U^{\wedge \{n\}}$ its renewal measure, an atom of mass 1 at the origin plus a density r . We use the following standard facts [8, Ch. XI].

Lemma 4.1. (a) $r(x) = \mathbb{1}_{\{[0,1)\}}(x) + \int_{\{ \max(0, x-1) \}^{\wedge \{x\}}} r(t) dt$. Hence $r(x) = e^{\wedge x}$ on $[0,1)$, $r(x) = e^{\wedge x} - x e^{\wedge(x-1)}$ on $[1,2)$, and $r(x) = \int_{\{x-1\}^{\wedge \{x\}}} r(t) dt$ for $x \geq 1$. (b) $g := r - 2$ decays exponentially: there are $C_0, \kappa > 0$ with $|g(x)| \leq C_0 e^{\{-\kappa x\}}$ for all $x \geq 0$; in particular $g \in L^1 \cap L^2$. (c) $G_1(t) := \int_0^{\wedge t} g \geq -1$ for every $t \geq 0$.

Proof. (a) is the renewal equation $r = U + U \star r$, solved on $[0,1)$ and $[1,2)$ by the method of steps. For (b), the Laplace transform of g is $s \hat{q}(s)/(s - 1 + e^{\wedge(-s)})$ with $q(x) = (2x - 1) \mathbb{1}_{\{[0,1)\}}(x)$. It is analytic in $\text{Re } s > -2.0888$: the non-zero roots of $s = 1 - e^{\wedge(-s)}$ all have real part at most -2.0888 , and the double zero of the denominator at $s = 0$ is cancelled because the numerator also vanishes to second order there, $\hat{q}(0) = \int q = 0$. Exponential decay of $r - 2$ then follows from the standard results on renewal measures of distributions that have a density and finite exponential moments [19, 20]; see also [8, Ch. XI]. For (c), $\int_0^{\wedge t} r$ is the renewal function $H(t)$, and Wald's identity gives $H(t) + 1 \geq t/E[U] = 2t$. ■

Lemma 4.2. Let $A(t) = \int_0^{\wedge \infty} g(x) g(x+t) dx$. Then $A(t) = e^{\wedge t} - 2t - 2/3$ for $0 \leq t \leq 1$. In particular $\int_0^{\wedge \infty} (r(x) - 2)^2 dx = 1/3$.

Proof. By Lemma 4.1(a), $g = U \star g + q$ with q as above, so in the sense of distributions on $(0, \infty)$, $g' = g - g(\cdot - 1) \mathbb{1}_{\{x \geq 1\}} + 2 \cdot \mathbb{1}_{\{(0,1)\}} - \delta_1$, where δ_1 records the jump of g by -1 at $x = 1$. For $0 < t < 1$, differentiating A under the integral — splitting at $x = 1 - t$, where $g(x+t)$ jumps — gives the functional equation

$$A'(t) = A(t) - A(1-t) + 2 \int_0^{\wedge \{1-t\}} g - g(1-t) = A(t) - A(1-t) + e^{\wedge \{1-t\}} + 4t - 4. \quad (4.1)$$

Put $D(t) = A(t) - A(1-t)$. Applying (4.1) at t and at $1-t$ gives $D'(t) = e^{\wedge t} + e^{\wedge \{1-t\}} - 4$ with $D(1/2) = 0$, so $D(t) = e^{\wedge t} - e^{\wedge \{1-t\}} - 4t + 2$. Then (4.1) reads $A'(t) = e^{\wedge t} - 2$, whence $A(t) = e^{\wedge t} - 2t + C$ on $[0,1]$. To fix C , note that for $t > 1$ the same differentiation gives $A'(t) = A(t) - A(t-1)$, the delay equation satisfied by r . For solutions of this equation, $\Phi(t) = A(t) -$

$\int_{-\{t-1\}^{\wedge}\{t\}} A$ and $\Psi(t) = \int_{-\{t-1\}^{\wedge}\{t\}} (s - t + 1) A(s) ds$ satisfy $\Phi' = 0$ and $\Psi' = \Phi$ on $(1, \infty)$. Since $g \in L^2$, $A(t) \rightarrow 0$ as $t \rightarrow \infty$, so $\Phi \equiv 0$ and then $\Psi \equiv 0$. Now $\Psi(1) = \int_0^1 s A(s) ds = 1/3 + C/2$, so $C = -2/3$, and $\int g^2 = A(0) = 1/3$. ■

Lemma 4.3 (the weight identity). *Let $\nu = 2M - 4 \cdot \text{Leb} = 2\delta_0 + 2g$, and for $s > 0$ put*

$$h(s) = 2s + \nu([0, s]) - \frac{1}{4} \int \nu(du) \nu(d(u + s)). \quad (4.2)$$

Then $h(s) = 8/3$ for $0 < s < 1$, and $h(s) \geq 8/3$ for $s \geq 1$.

Proof. Here $\nu([0, s]) = 2 + 2G_1(s)$, and for $s > 0$ the last integral is $4g(s) + 4A(s)$: an atom-density term plus a density-density term. Hence

$$h(s) = 2s + 2 + 2G_1(s) - g(s) - A(s). \quad (4.3)$$

For $0 < s < 1$, Lemma 4.1(a) and Lemma 4.2 give $g(s) = e^s - 2$, $G_1(s) = e^s - 1 - 2s$ and $A(s) = e^s - 2s - 2/3$. Substituting, every term cancels except $8/3$. For $s \geq 1$, Lemma 4.1(a) gives $g(s) = G_1(s) - G_1(s - 1)$, so $h(s) = 2s + 2 + G_1(s) + G_1(s - 1) - A(s)$. By Cauchy-Schwarz, $A(s) \leq A(0) = 1/3$. If $s \geq 3/2$, Lemma 4.1(c) gives $h(s) \geq 2s - 1/3 \geq 8/3$. If $1 \leq s < 3/2$, two more facts are needed. First, on $[1, 3/2]$ we have $r' = e^{\{x-1\}}(e - x - 1) > 0$ and $r(1+) = e - 1$, so $g \geq e - 3$ there and $G_1(s) \geq (3/2)(e - 3)$. Second, G_1 is decreasing on $[0, \ln 2] \supset [0, 1/2]$, so $G_1(s - 1) \geq e^{\{1/2\}} - 2$. Together these give $h(s) \geq 4 + (3/2)(e - 3) + e^{\{1/2\}} - 2 - 1/3 = 2.8928... > 8/3$. ■

Figure 1. The weight (4.2) that the renewal multiplier μ_1 places on the kernels' autocorrelation: exactly $8/3$ on the first window, larger beyond it. Kernels on $[0, 1]$ only see lags $|s| \leq 1$, which the choice $w = 4/(3a)$ keeps inside the first window whenever $a \leq 4/3$.

4.2 The lower bound

For $w > 0$ let $\mu_w = 2w \cdot M_w$ as in (1.1). Equivalently $\mu_w(du) = 2w\delta_0 + 2r(u/w) du$, and $\nu_w = \mu_w - 4 \cdot \text{Leb}$ is a finite signed measure by Lemma 4.1(b).

Theorem 4.4. *Every continuum certificate satisfies $b \geq 4w/3 - (w^2/2) \cdot a$ for every $w > 0$. Consequently $a \cdot b \geq 8/9$.*

Proof. Apply Lemma 3.1 with $\mu = \mu_w$. Since R is even, $\int T d\nu = \int_0^\infty R(s) \nu([0, s]) ds$ by Fubini, and $\int \int R(u - v) \nu(du) \nu(dv) = (2w)^2 R(0) + 2 \int_0^\infty R(s) \int \nu(du) \nu(d(u+s)) ds$. The right-hand side of (3.2) is therefore

$$\int_0^\infty R(s) [2s + \nu_w([0, s]) - \frac{1}{4} \int \nu_w(du) \nu_w(d(u+s))] ds - (w^2/2) \cdot R(0).$$

By scaling, the bracket equals $w \cdot h(s/w)$, with h as in (4.2), and Lemma 4.3 gives $w \cdot h(s/w) \geq 8w/3$ for every $s > 0$. Since $R \geq 0$ and $\int_0^\infty R = 1/2$, the bound is at least $(8w/3) \cdot 1/2 - (w^2/2) \cdot a$. Taking $w = 4/(3a)$ gives $b \geq 8/(9a)$. ■

Corollary 4.5 (Theorem 1(a)). *Every certificate, discrete or continuum, symmetric or two-sided, with any number of kernels, satisfies $\gamma = \sqrt{ab} \geq 2\sqrt{2}/3$. This follows from Theorem 4.4 and Lemma 2.5. The positivity hypothesis $b > 0$ of Propositions 2.1 and 2.3 is automatic, since $b \geq 8/(9a) > 0$.*

Three features of the argument stand out. The multiplier depends on the certificate only through a , exactly as the atom $2/a$ of the one-atom bound did. Symmetry is never used, and mixtures are handled through R , with no reduction to single kernels. And when $a \leq 4/3$, the width $w = 4/(3a)$ is at least 1, so the kernels see only the first window, where the weight is constant and (3.2) with μ_w equals $4w/3 - (w^2/2)a$ exactly, for every kernel: the renewal multiplier cannot distinguish kernels of equal energy.

5. Sharpness

Theorem 5.1. *Let $\rho(x) = 2x$ on $[0,1]$. At the end whose covering condition involves the reflected kernel $\bar{\rho}(x) = 2(1-x)$ take $\omega' \equiv 1$. At the other end take $\omega = \frac{1}{4}(\mu_1 * \rho)$, that is, $\omega(t) = e^{-t}$ for $0 \leq t < 1$ and $\omega(t) = \int_0^1 x \cdot r(t-x) dx$ for $t \geq 1$. This is a continuum certificate with $a = 4/3$ and $b = 2/3$; hence $a \cdot b = 8/9$ and $\gamma = 2\sqrt{2}/3$.*

Proof. The autocorrelation of ρ is $R(s) = (2/3)(1-|s|)^2(2+|s|)$. A direct computation on $0 \leq u \leq 1$, using $r(v) = e^{-v}$ on $[0,1]$ and $r(v) = e^{-v} - v e^{-v-1}$ on $[1,2]$, gives $(\mu_1 * R)(u) = 4$. The renewal identity $\mu_1 = 2\delta_0 + U * \mu_1$ then gives $(\mu_1 * R)(u) = \int_{-\infty}^{\{u-1\}} (\mu_1 * R)$ for $u \geq 1$, since R vanishes there, so $(\mu_1 * R) \equiv 4$ on $[0, \infty)$. As $\int \rho(x)\omega(u+x) dx = \frac{1}{4}(\mu_1 * R)(u)$, the covering condition holds with equality everywhere at that end. At the other end, $\frac{1}{4}(\mu_1 * \bar{\rho}) \equiv 1$ by the same kind of computation (for $0 \leq t < 1$ it is $(1-t) + \int_0^t e^{-v}(1-t+v) dv = 1$), so $\omega' \equiv 1$ is both feasible and of the form required in Lemma 3.1, with equality in its covering condition. The equality conditions of Lemma 3.1 therefore hold, so b equals the right-hand side of (3.2) with $\mu = \mu_1$. Since $a = \int_0^1 (2x)^2 = 4/3$ and R lives inside the first window of μ_1 , that value is exactly $4/3 - \frac{1}{2} \cdot (4/3) = 2/3$, as computed in the proof of Theorem 4.4. Finally $\omega - 1$ decays exponentially, by Lemma 4.1(b). ■

Proposition 5.2 (Theorem 1(b)). *For every $m \geq 2$ there is a two-sided certificate with one kernel at resolution m and boundary depth $L = \max(2, \lceil (2/\kappa) \log m \rceil)$, with $a = 4/3 - 1/(3m^2)$ and $a \cdot b \leq 8/9 + C(\log m)/m^2$, where C depends only on the constants C_0, κ of Lemma 4.1(b). Hence $\gamma^* = 2\sqrt{2}/3$.*

The proof, in Appendix B, discretizes the certificate of Theorem 5.1 by cell averages and repairs the covering deficits with a uniform increment η on the boundary vector. The one point that needs care is the end of the boundary box. At the last shifts only a few kernel cells fall inside the box — at the very last shift only the first, of mass $1/m^2$ — so the

increment raises the covering sum there only in proportion to that in-box mass. The deficits at those shifts are proportional to the same mass, because every in-box cell there sits at depth at least $L - 1$, where the profile is exponentially close to 1; so an increment of size $O(m^{-2})$ suffices.

With the exactly optimal boundary vectors instead of cell averages, the discrete values behave as cleanly as one could hope: with $L = 8$, $b = 2/3 + 1/(3m^2)$ to within 3×10^{-8} for $m \leq 512$ (Table 2), so $a \cdot b = 8/9 + 2/(9m^2) - 1/(9m^4)$ to that accuracy. The certificate at $m = 512$ is the one of Theorem 2.

m	γ	$b - 2/3$	$1/(3m^2)$
32	0.9429240903	3.2555×10^{-4}	3.2552×10^{-4}
64	0.9428378328	8.1413×10^{-5}	8.1380×10^{-5}
128	0.9428162572	2.0377×10^{-5}	2.0345×10^{-5}
256	0.9428108625	5.1183×10^{-6}	5.0863×10^{-6}
512	0.9428095137	1.3035×10^{-6}	1.2716×10^{-6}

Table 2. The discretized linear kernel at $L = 8$ with exactly optimal boundary vectors at both ends. The certificates at $m = 256$ and $m = 512$ were verified in exact rational arithmetic (4098 and 8194 inequalities), giving $\gamma < 0.942810871$ and $\gamma < 0.942809522$.

Figure 2. Left: the optimal dual multiplier at numerically optimal two-sided certificates ($m = 64$ and 256 ; density divided by the flat value 4, atom not shown), against twice the renewal density of the uniform distribution. Right: the boundary profiles of the extremal certificate of Theorem 5.1.

6. The numerical picture

The results above were found numerically, and the numerics remain a useful consistency check. All values below come from exact quadratic-program solves for fixed kernels, nested in an L-BFGS optimization over kernels and mixing weights. Where marked, certificates were verified in exact rational arithmetic.

6.1 Resolution scans

With the boundary depth fixed at $L = 4$, optimized symmetric and two-sided certificates with two kernels give the values in Table 3. Both scans converge in $1/m^2$ and extrapolate to 0.9428091, about 4×10^{-8} above $2\sqrt{2}/3$. The difference is the cost of the four-width box: the extremal certificate needs a boundary layer about six kernel widths deep, and at $L = 4$ the linear kernel loses about 10^{-4} in b , which optimized mixtures recover almost entirely. A two-sided certificate at resolution m does about as well as a symmetric one at $2m$.

m	symmetric, $L = 4$	two-sided, $L = 4$	linear kernel, $L = 8$
32	0.94324229	0.9429206688	0.9429240903
64	0.94292072	0.9428374024	0.9428378328
128	0.94283748	0.9428162103	0.9428162572
256	0.94281628	0.9428108643	0.9428108625
512	0.9428108915	—	0.9428095137

Table 3. Optimized certificates (two kernels, $L = 4$) against the linear kernel at $L = 8$. The two-sided entries and the symmetric entry at $m = 512$ are verified in exact arithmetic; the linear-kernel entries are exact quadratic-program values. Richardson extrapolation gives 0.9428091 for both $L = 4$ scans and 0.94280906 for the linear kernel, against $2\sqrt{2}/3 = 0.94280904$.

Figure 3. The values of Table 3 against $1/m^2$, with $2\sqrt{2}/3$ marked.

6.2 The optimal multiplier

The dual solution of the boundary program at the optimal certificates converges, at rate $1/m$, to twice the renewal measure of $U(0,1)$ (Figure 2, left). At resolution m the atom has mass $2 - 2/m$, and the mean deviation of the density on $[0,1)$ is 0.010, 0.005 and 0.003 at $m = 64, 128$ and 256 . With every covering constraint active, the optimality conditions say that μ solves the Wiener–Hopf equation $(\mu \star R)(u) = 4$ for $u \geq 0$. Given $\mu = \mu_1$, that equation determines R on $[0,1]$ as the autocorrelation of $2x$, which is how the extremal kernel was found.

6.3 A family of multipliers

Before Theorem 4.4 was known, we tested a three-parameter family $\mu = 4 \cdot \text{Leb} + \beta \delta_0 + \theta(2r(u/w) - 4)$, with (w, β, θ) fitted per certificate under the constraint $\mu \geq 0$. We searched for the kernel minimizing the certified value over single kernels and two-kernel mixtures, at $m = 32, 64, 128$ and 256 . The worst certified $a \cdot b$ fell to $0.8890087, 0.8889188, 0.88889636$ and 0.88889075 , with drop ratios 4.006 and 4.000 , and extrapolated to 0.888888880 . The minimizers were not the linear kernel but mildly U-shaped kernels, at which the family selected $\beta = 2w, \theta = 1$ and $a \cdot w = 4/3$. Theorem 4.4 explains this. That choice reproduces exactly the multiplier of the proof, and the fitted family therefore never certifies less than $8/9$, with equality wherever that choice is also the best in the family.

6.4 The symmetric framework

Symmetric certificates cannot contain the extremal kernel, which is asymmetric. Theorem 1(a) bounds them below by $2\sqrt{2}/3$, and the best we have found gives $\gamma < 0.942810893$ (verified; $m = 512, L = 4$). Their scan at $L = 4$ extrapolates to the same 0.9428091 as the two-sided scan, and deepening the boundary layer from $L = 4$ to $L = 8$ at $m = 32$ improves the optimum by only 6×10^{-8} . Equality $\gamma^*_{\text{sym}} = 2\sqrt{2}/3$ would follow if the autocorrelation of $2x$ lay in the closed convex hull of autocorrelations of symmetric probability densities on $[0,1]$, and we expect the converse to hold as well. We leave the question open.

7. Keeping the variance term: the pair-level relaxation

The smoothing framework discards two quantities, the variance term in Cilleruelo's identity [5] and the slack in the bound $d(h) \leq 1$ (Section 8). A natural question is whether the method has an extension that keeps the variance term, and what the corresponding dual problem looks like. This section answers that question at the first natural level.

Proposition 7.1 rewrites the proof of Lemma C.1 as an identity in which both discarded quantities appear explicitly. The natural extension that keeps them is the level-one moment relaxation of the Sidon condition. It is a semidefinite program whose value $\hat{h}_2(N)$ is at least $h(N)$ (Definition 7.2). Every certificate bounds $\hat{h}_2(N)$ exactly as it bounds $h(N)$ (Proposition 7.3). Explicit feasible points show that nothing more can be gained: at $N = 10^4, 10^5, 10^6$ and 10^7 , $\hat{h}_2(N)$ lies within about 1 of $\sqrt{N} + (2\sqrt{2}/3)N^{1/4}$ (Theorem 7.6).

Keeping the variance term therefore gains at most $O(1)$, unless the argument uses more than the one- and two-point statistics of the set. Section 7.5 collects evidence that three-point information does not help either. Four-point information, the native form of the

Sidon condition, captures only a small and shrinking part of the relevant arithmetic obstruction.

Throughout, as in Appendix C, $A \subset \{0, \dots, N-1\}$ is a Sidon set with k elements, a two-sided certificate at resolution (m, L, K) is fixed, and $H = mh$ with $N \geq 2LH$. The objects K_r, S_r, Q_r, C_r, J and $c(x)$ are those of the proof of Lemma C.1.

7.1 The identity behind (C.1)

Proposition 7.1. *Put*

$$c(A) = \sum_{\{x \in A\}} c(x), \Delta(A) = \sum_r \lambda_r \sum_{\{d \neq 0, d \notin A - A\}} C_r(d),$$

$$V(A) = (\sum_r \lambda_r \|Q_r\|^2)(\sum_r \lambda_r \|S_r\|^2) - c(A)^2.$$

Then $c(A) \geq k$, $V(A) \geq 0$, $\Delta(A) \geq 0$, and

$$c(A)^2 + V(A) + (N - 1 + bH) \cdot \Delta(A) = (N - 1 + bH) \cdot (1 + (k - 1)a/H). \quad (7.1)$$

Proof. The inequality $c(A) \geq k$ is (C.4) summed over A . Exchanging sums gives $c(A) = \sum_r \lambda_r \langle Q_r, S_r \rangle$, so $V(A) \geq 0$ is the Cauchy–Schwarz inequality of Lemma C.1. Since A is Sidon, each nonzero difference occurs once in $A - A$. Hence $\|S_r\|^2 = kC_r(0) + \sum_{\{d \in A - A, d \neq 0\}} C_r(d) = kC_r(0) + 1 - C_r(0) - \Delta_r$, where Δ_r is the inner sum in $\Delta(A)$. Averaging gives $\sum_r \lambda_r \|S_r\|^2 = 1 + (k - 1)a/H - \Delta(A)$. Multiplying by (C.3) and inserting the definition of $V(A)$ gives (7.1). ■

The proof of Lemma C.1 discards exactly three quantities: the covering slack $c(A)^2 - k^2$, the Cauchy–Schwarz defect $V(A)$, and the difference slack $(N - 1 + bH)\Delta(A)$.

Relation to Cilleruelo’s identity. Take the flat kernel with $u = v \equiv 1$, a certificate with $a = b = 1$ and $c \equiv 1$. Then $c(A) = k$, $Q \equiv 1$, and $H \cdot S(x)$ is the number of elements of A in the window $\{x - H + 1, \dots, x\}$. So $V(A)/(N + H - 1) = \sum_{\{x \in J\}} (S(x) - k/(N + H - 1))^2$ is the variance of the window counts divided by H^2 . Likewise $H^2 \cdot \Delta(A) = \sum_{\{0 < |d| < H, d \notin A - A\}} (H - |d|)$ is the slack in $d(h) \leq 1$. In this case (7.1) is, up to normalization, the identity of Cilleruelo [5]. In general, $V(A)/\sum_r \lambda_r \|Q_r\|^2$ is the squared distance, in the weighted direct sum, of the smoothed set (S_r) from the line spanned by the weights (Q_r) .

The two discarded quantities are coupled. For fixed k and $c(A)$, identity (7.1) fixes $V(A) + (N - 1 + bH)\Delta(A)$. A set that uses fewer small differences has a larger variance, and conversely. A lower bound for either quantity alone is therefore equivalent to an upper bound on k . Keeping the variance term can help only through information the certificate does not already use.

A normalization. If the points of A were independent with density k/N , the expected value of $V(A)/(N - 1 + bH)$ would be close to $k \cdot a/H$, the diagonal term of the energy. Put $\phi(A) = V(A) \cdot H / ((N - 1 + bH) \cdot k \cdot a)$, so that $\phi \approx 1$ corresponds to Poisson-like fluctuations. Suppose $k \geq \sqrt{N} + \gamma N^{1/4}$ and $H = tN^{3/4}$. Then (7.1), with $c(A) \geq k$ and $\Delta(A) \geq 0$, gives

$$\phi(A) \leq 1 + t(bt - 2\gamma)/a + O(N^{-1/4}). \quad (7.2)$$

At the certificate's own scale $t = \sqrt{a/b}$, the right-hand side is $2(1 - \gamma/\sqrt{ab}) + O(N^{-1/4})$. So every Sidon set more than halfway to the bound of a certificate has sub-Poissonian fluctuations, $\phi < 1$, at scale $\sqrt{a/b} \cdot N^{3/4}$. This is the structural information carried by the variance term, in the spirit of Cilleruelo's use of it [5].

We measured ϕ with a discretized linear-kernel certificate. The optimal Golomb rulers with 12 to 20 marks have ϕ between 0.23 and 0.45, and most of them use every difference below H . The full-period sets of Bose [3] and Ruzsa [16] have ϕ between 1.1 and 1.5. A uniformly random subset of the same size and length gives 1.57.

7.2 The pair-level relaxation

Definition 7.2. For $N \geq 2$ let $\hat{h}_2(N)$ be the value of the semidefinite program maximize $\sum_x f(x)$ over $f \in \mathbb{R}^N$ and symmetric $\Gamma \in \mathbb{R}^{(N \times N)}$, subject to (7.3)

(i) $[[1, f^T], [f, \Gamma]] \geq 0$; (ii) $\Gamma(x, x) = f(x)$; (iii) $0 \leq f \leq 1, \Gamma \geq 0$ entrywise;

(iv) $\sum_x \Gamma(x, x + d) \leq 1$ for $1 \leq d \leq N - 1$,

with indices in $\{0, \dots, N - 1\}$.

For a Sidon set A , the pair $f = \mathbb{1}_A, \Gamma = \mathbb{1}_A \mathbb{1}_A^T$ is feasible with value k , so $h(N) \leq \hat{h}_2(N)$.

Conditions (i)–(iii) are the level-one moment conditions of Lasserre [11] and the pair conditions of Sherali and Adams [17] for 0/1 variables. Condition (iv) is the Sidon condition in expectation: at most one pair with each difference. The matrix $\Gamma - ff^T$ is the covariance of the set. Condition (i) keeps it positive semidefinite instead of discarding it, and (iv) keeps each difference budget separately. Both quantities of Proposition 7.1 thus become linear functionals of (f, Γ) , and the proof of Lemma C.1 becomes a dual solution. In this sense $\hat{h}_2(N)$ is the natural extension of the smoothing argument that keeps them, at the level of pair statistics.

Proposition 7.3. For every two-sided certificate and every $H = mh$ with $N \geq 2LH$,

$$\hat{h}_2(N)^2 \leq (N - 1 + bH) \cdot (1 + (\hat{h}_2(N) - 1)a/H). \quad (7.4)$$

Consequently (C.5) holds with $h(N)$ replaced by $\hat{h}_2(N)$, and

$$\limsup_{\{N \rightarrow \infty\}} (\hat{h}_2(N) - \sqrt{N})/N^{1/4} \leq 2\sqrt{2}/3. \quad (7.5)$$

Proof. Let (f, Γ) be feasible and put $\kappa = \sum_x f(x)$.

Covering. Since $f \geq 0$, summing (C.4) against f gives $\kappa \leq \sum_x f(x)c(x) = \sum_r \lambda_r \langle Q_r, K_r \star f \rangle$, where $(K_r \star f)(z) = \sum_y K_r(z - y) f(y)$ is supported in J . By Cauchy–Schwarz in the weighted direct sum, $\kappa^2 \leq (\sum_r \lambda_r \|Q_r\|^2)(\sum_r \lambda_r \|K_r \star f\|^2)$.

Energy. Write $\|K_r \star f\|^2 = f^T \tilde{\Gamma}_r f$, where $\tilde{\Gamma}_r(x, y) = C_r(x - y)$ is a Gram matrix and hence positive semidefinite. By (i), $\Gamma - ff^T \geq 0$, so

$$f^T \tilde{\Gamma}_r f \leq \text{tr}(\tilde{\Gamma}_r \Gamma) = C_r(0)\kappa + \sum_{\{d \neq 0\}} C_r(d) \sum_x \Gamma(x, x + d) \leq C_r(0)\kappa + 1 - C_r(0).$$

The equality uses (ii), and the last step uses $C_r \geq 0$ and (iv); for $d < 0$ use the symmetry of Γ . Averaging over r and using (C.3) gives (7.4).

Consequences. The proof of Corollary C.2 uses only (C.1) and the bound $k \leq \sqrt{2N} + 1$. The analogue $\kappa \leq \sqrt{2N} + 1$ follows from $\kappa^2 \leq 1^T \Gamma 1 \leq \kappa + 2(N - 1)$, by (i), (ii) and (iv). The certificates of Appendix B have $a \cdot b \leq 8/9 + C(\log m)/m^2$, which gives (7.5). ■

Only (i), (ii), (iv) and $f \geq 0$ were used, so the bound also holds without $\Gamma \geq 0$ and $f \leq 1$. The same argument applies to any nonnegative kernel K of total mass 1 together with any weight function Q on J satisfying (C.4) at every $x \in \{0, \dots, N - 1\}$; it gives $\hat{h}_2(N)^2 \leq \|Q\|^2(1 + (\hat{h}_2(N) - 1)\sum_t K(t)^2)$. Section 7.4 uses this finite- N form as a comparison bound, with the linear kernel and the exactly optimal Q .

7.3 The relaxation reaches $2\sqrt{2}/3$

The lower bound comes from explicit feasible points whose shape is read off from the equality conditions of Lemma 3.1. A set that makes the Cauchy–Schwarz step of Lemma C.1 tight must have smoothed density proportional to the weights. By Lemma 3.1, near each end its density must then follow $\frac{1}{4}\mu_1 = \frac{1}{2}\delta_0 + \frac{1}{2}r$. That is an atom, followed by a density that starts at half the bulk value, rises to $e/2$ at one kernel width and settles to the bulk value. The first ingredient is that this profile is invisible to short-range pair counts, which is another reading of Lemma 4.3.

Lemma 7.4. Let $\sigma = \frac{1}{2}\delta_0 + \frac{1}{2}r \cdot \text{Leb}$ on $[0, \infty)$, and let $\tau = \sigma - \text{Leb}|_{[0, \infty)}$, a finite signed measure with $\tau([0, \infty)) = 1/3$. For $s > 0$ put

$$\mathcal{B}(s) = 2\tau([0, \infty)) - \tau([0, s]) + \int \tau(du) \tau(d(u + s)). \quad (7.6)$$

Then $\mathcal{B}(s) = s/2 + (8/3 - h(s))/4$, with h as in (4.2). In particular $\mathcal{B}(s) = s/2$ for $0 < s < 1$, and $\mathcal{B}(s) \leq s/2$ for $s \geq 1$.

Proof. We have $\tau = \frac{1}{2}\delta_0 + \frac{1}{2}g \cdot \text{Leb}$. The Laplace transform of g in the proof of Lemma 4.1(b) is $s\hat{q}(s)/(s-1+e^{-s})$, and $\hat{q}(s) = -s/6 + O(s^2)$, $s-1+e^{-s} = s^2/2 + O(s^3)$; it therefore takes the value $-1/3$ at the origin, so $\int_0^\infty g = -1/3$ and $\tau([0, \infty)) = \frac{1}{2} - 1/6 = 1/3$. Next, $\tau([0, s]) = \frac{1}{2} + \frac{1}{2}G_1(s)$. For $s > 0$ the density-atom term of the last integral vanishes, so $\int \tau(du)\tau(d(u+s)) = \frac{1}{4}(g(s) + A(s))$. Hence

$$\mathcal{B}(s) = 1/6 - \frac{1}{2}G_1(s) + \frac{1}{4}(g(s) + A(s)),$$

and substituting (4.3) gives $\mathcal{B}(s) = s/2 + 2/3 - h(s)/4$. Lemma 4.3 finishes the proof. ■

What the lemma says. Take a density on $[0, \mathcal{L}]$ that equals σ at both ends, with lengths in units of the kernel width. Its autocorrelation at a lag $s \in (0, 1)$ is $\mathcal{L} - s$ from the bulk plus $2\mathcal{B}(s)$ from the two ends, up to terms exponentially small in $\mathcal{L}$ (Lemma 4.1(b)). By Lemma 7.4 this is exactly $\mathcal{L}$: the two ends supply exactly the pairs that the finite length removes. On a grid of step $1/2000$ the numerical autocorrelation is flat to 3×10^{-4} on $(0, 1)$ and strictly below $\mathcal{L}$ beyond.

So Lemma 4.3 says that the boundary layer of the extremal certificate is invisible to short-range pair counts. This is a reading of the lemma, not a new proof. Whether it leads to the conceptual proof asked for in Section 8, we leave open.

Lemma 7.5. Let $0 < \delta < 1$ and let ℓ be an integer with $1 \leq \ell \leq 1/\delta$. Let Φ_ℓ be the $N \times N$ Toeplitz matrix with entries $(1 - |x - y|/\ell)_+$. Let $f \in [0, 1]^N$ and $\tilde{\psi} \in \mathbb{R}^N$ satisfy $|\tilde{\psi}(x)| \leq f(x)$ and $(1 - \delta)\tilde{\psi}(x)^2 \leq \delta(f(x) - f(x)^2)$ for every x . Put

$$\Gamma = ff^\top + D(I - \delta\Phi_\ell)D + \text{diag}(\varepsilon), \quad D = \text{diag}(\tilde{\psi}/\sqrt{\delta}), \quad \varepsilon(x) = f(x) - f(x)^2 - (1 - \delta)\tilde{\psi}(x)^2/\delta. \quad (7.7)$$

Then (f, Γ) satisfies (i)–(iii). Moreover, for $d \geq 1$,

$$\sum_x \Gamma(x, x+d) = \sum_x f(x)f(x+d) - (1 - d/\ell)_+ \cdot \sum_x \tilde{\psi}(x)\tilde{\psi}(x+d). \quad (7.8)$$

Proof. Φ_ℓ is a finite section of the Toeplitz operator whose symbol is the Fejér kernel $(1/\ell)|\sum_{|j|<\ell} e^{ij\theta}|^2$, which takes values in $[0, \ell]$. Hence $0 \leq \Phi_\ell \leq \ell I \leq \delta^{-1}I$, so $I - \delta\Phi_\ell \geq 0$. It follows that $\Gamma - ff^\top \geq 0$, since $\varepsilon \geq 0$ by hypothesis, and this gives (i) by the Schur complement.

The diagonal of $D(I - \delta\Phi_\ell)D$ is $(1 - \delta)\tilde{\psi}^2/\delta$, which gives (ii). Off the diagonal, $\Gamma(x, y) = f(x)f(y) - \tilde{\psi}(x)\tilde{\psi}(y)\Phi_\ell(x, y) \geq f(x)f(y) - |\tilde{\psi}(x)\tilde{\psi}(y)| \geq 0$, which gives (iii). Summing along diagonals gives (7.8). ■

The term $D(I - \delta\Phi_\ell)D$ is a covariance with a triangular hole: each bulk point gives up about one partner at lags below ℓ . Taking $\ell = \lfloor 1/\delta \rfloor$, as large as the Fejér bound allows, makes this hole integrate to one partner per point, so the bulk is *hyperuniform* in the sense of Torquato and Stillinger [21].

The construction. Fix a bulk density δ , a scale $H = t \cdot N^{3/4}$, an atom fraction $\beta \in (0, 1]$, and $\ell = \lfloor 1/\delta \rfloor$. Let $\sigma_c(u) = \frac{1}{2}r(u)$ for $u \geq 0$ and $\sigma_c(u) = 0$ for $u < 0$. We use a numerical solution of the renewal equation and set $\sigma_c(u) = 1$ for $u \geq 13.9$; feasibility does not depend on the accuracy of this approximation, since (iv) is checked for the profile actually used. Put

$$\psi(x) = \delta \cdot [\sigma_c((x + \frac{1}{2} - \ell/2)/H) + \sigma_c((N - \frac{1}{2} - x - \ell/2)/H) - 1],$$

$$\alpha(x) = \beta \delta H / (2\ell) \text{ for } x < \ell \text{ or } x \geq N - \ell, \alpha(x) = 0 \text{ otherwise, (7.9)}$$

$$f = \psi + \alpha, \tilde{\psi} = \min(\psi, \sqrt{\delta(f - f^2)/(1 - \delta)}),$$

and let Γ be given by (7.7).

Here ψ is the continuous part $\frac{1}{2}r$ of $\frac{1}{4}\mu_1$ at each end, started at the centre of the atom, and α is its atom $\frac{1}{2}\delta_0$, spread uniformly over the first and last ℓ sites. Whenever $\psi \geq 0$, as it is in all our computations, the hypotheses of Lemma 7.5 hold by construction. The atoms are independent of each other and of the bulk; only the bulk carries the hole.

Why the scale is $\sqrt{2} \cdot N^{3/4}$. Each atom has about $(\frac{1}{2}\delta H)^2$ ordered pairs at lags below ℓ , with the same triangular profile $(1 - d/\ell)_+$ as the hole. By (7.8) and Lemma 7.4, condition (iv) at a lag $d < \ell$ reads, to leading order,

$$\delta^2 N + (1 - d/\ell) \cdot [2(\frac{1}{2}\delta H)^2/\ell - \delta^2 N] \leq 1.$$

The atoms fill the vacated lags exactly when $2(\frac{1}{2}\delta H)^2 = \ell \approx \sqrt{N}$, that is, when $H = \sqrt{2} \cdot N^{3/4}$. This is the certificate's optimal scale $\sqrt{(a/b)} \cdot N^{3/4}$ for $a = 4/3$ and $b = 2/3$, and the balance holds at no other scale.

The count. At that scale $\delta^2 N \rightarrow 1$. Each end carries an excess of $b/2 = 1/3$ in units of δH : the atom contributes $\frac{1}{2}$ and the continuous part $\frac{1}{2} \int_0^\infty g = -1/6$. So $\sum_x f(x) = \delta N + (2/3)\delta H + O(1) = \sqrt{N} + (2\sqrt{2/3})N^{1/4} + O(1)$.

The computation. For given N , t and β , we took the largest δ satisfying (iv) at every d , found by bisection. The difference sums (7.8) were evaluated for all d by fast Fourier transform, and t and β were optimized. Table 4 gives the results.

N	t	β	$\delta\sqrt{N}$	$\sum_x f(x)$	$(\sum f - \sqrt{N})/N^{1/4}$	$\sqrt{N} + (2\sqrt{2/3})N^{1/4} - \sum f$
10^4	1.344	0.9936	1.00514	108.441	0.844	0.99
10^5	1.414	0.9698	1.00221	331.969	0.885	1.02
10^6	1.414	0.9837	1.00069	1028.794	0.911	1.02
10^7	1.414	0.9912	1.00021	3214.284	0.925	1.01

Table 4. Explicit feasible points of (7.3). The shortfall from $\sqrt{N} + (2\sqrt{2}/3)N^{1/4}$ in the last column is constant to within 0.04 over three decades. At $N = 10^7$, β was extrapolated from the smaller cases rather than optimized.

Theorem 7.6 (computational). *For $N \in \{10^4, 10^5, 10^6, 10^7\}$, the points (7.7), (7.9) with the parameters of Table 4 are feasible for (7.3), and*

$$\hat{h}_2(N) \geq \sqrt{N} + (2\sqrt{2}/3)N^{1/4} - 1.03. \quad (7.10)$$

Conditions (i)–(iii) hold by Lemma 7.5, whose hypotheses were checked on the computed vectors. In all four cases $\psi \geq 0$ and $f \leq 0.073$. The function ε vanishes identically in the bulk, and its computed values are at least -4×10^{-18} , which is rounding.

Condition (iv) was evaluated in double precision for every d . The largest difference sum at the four sizes is at most $1 - 1.3 \times 10^{-8}$. Lowering δ by a relative 10^{-7} leaves every difference sum at most $1 - 2 \times 10^{-7}$, far above any rounding error, and changes Σf by at most 3.3×10^{-4} .

As an end-to-end check we formed a feasible point of the construction explicitly at $N = 3000$. Its moment matrix has smallest eigenvalue 6.1×10^{-5} , $\text{diag}(\Gamma) = f$ to 10^{-17} and $\Gamma \geq 0$, and direct summation of (iv) agrees with the fast Fourier transform to 4×10^{-16} .

Conjecture 7.7. $\hat{h}_2(N) = \sqrt{N} + (2\sqrt{2}/3)N^{1/4} + O(1)$. In particular, $\lim_{N \rightarrow \infty} (\hat{h}_2(N) - \sqrt{N})/N^{1/4} = 2\sqrt{2}/3$.

The upper half is Proposition 7.3. For the lower half, the construction satisfies (iv) up to four errors: the finite width of the atoms, the reduction $\tilde{\psi} < \psi$ where the profile exceeds the bulk density, atom–bulk pairs at short lags, and discretization. Each has relative size $O(N^{-1/4})$ in the budget of a single lag. They force $\beta = 1 - O(N^{-1/4})$, and we observe $1 - \beta \approx 0.030, 0.016$ and 0.009 at $N = 10^5, 10^6$ and 10^7 ; they cost $O(1)$ in Σf . Writing out this error analysis would prove the conjecture. We have not done so.

7.4 Structure of the optimum at moderate N

We solved (7.3) for $20 \leq N \leq 180$ with Clarabel [9] through CVXPY [6], after block-diagonalizing by the reflection $x \mapsto N - 1 - x$. Table 5 compares the optimum with $h(N)$ and with the finite- N smoothing bound $s(N)$ of Section 7.2 for the linear kernel, minimized over H .

N	$h(N)$	$\hat{h}_2(N)$	$s(N)$	$s(N) - \hat{h}_2(N)$
20	6	6.4291	6.5822	0.1531
32	7	7.9189	8.0726	0.1537

N	h(N)	$\hat{h}_2(N)$	s(N)	s(N) - $\hat{h}_2(N)$
40	8	8.7405	8.8939	0.1534
50	9	9.6469	9.7993	0.1524
64	10	10.7568	10.9089	0.1521
80	11	11.8671	12.0198	0.1527
100	12	13.0938	13.2489	0.1551
120	13	14.1959	14.3522	0.1563
140	14	15.2037	15.3612	0.1575
160	15	16.1372	16.2958	0.1586
180	16	17.0104	17.1699	0.1595

Table 5. The pair-level relaxation against the smoothing bound. The gap stays between 0.152 and 0.160 while N grows ninefold; a difference of 0.05 in the second-order constant would open it by about 0.08 over this range.

Two warnings on reading Table 5. First, second-order constants cannot be fitted reliably from $N \leq 180$. Fitting $\sqrt{N} + \gamma N^{1/4} + c_0 + c_1 N^{-1/4}$ to $s(N)$ itself, whose limit is 0.943, returns anything from 0.85 to 0.96. The asymptotic claim therefore rests on Section 7.3, not on these fits. Second, Clarabel flagged some instances as possibly inaccurate. The returned solutions satisfy every constraint of (7.3) to within 10^{-9} , and re-solving without the symmetry reduction gives identical values at $N = 20, 40$ and 64 .

The optimal dual is the smoothing certificate. Denote by y_d the optimal multipliers of the difference constraints (iv). They fit $y_d \approx c \cdot R(d/H_{\text{fit}})$, with $R(s) = (2/3)(1 - |s|)^2(2 + |s|)$ the autocorrelation of $2x$ (Theorem 5.1). The relative L^2 misfit is 6.1%, 4.1%, 3.3%, 2.9%, 2.6% and 2.5% at $N = 32, 50, 64, 80, 100$ and 120 . The fitted scale H_{fit} is 22.2, 30.7, 35.3, 39.6, 47.7 and 55.6, against 22, 30, 35, 41, 49 and 56 for the optimal H of $s(N)$. The only visible departure is at $d = 1$. The flat kernel of Lindström fits more than twice as badly (5.8% at $N = 120$).

The optimal primal is the configuration of Section 7.3. At $N = 120$:

- *Endpoints and atoms.* The endpoints have $f = 0.98$. The first six sites at each end carry total mass 2.49, against $N^{1/4}/\sqrt{2} = 2.34$ predicted within $\sqrt{N}/2 \approx 5.5$ sites. A depleted stretch follows.

- *Bulk hard core.* In the middle half, $\Gamma(x, x + d) = 0$ for $1 \leq d \leq 6$, and $\Gamma(x, x + d) \approx f(x)f(x + d)$ from $d = 8$ on; the predicted radius is $\sqrt{N}/2 \approx 5.5$.
- *Hyperuniformity.* The covariance hole sums to about -1 per point.
- *Difference budgets.* Condition (iv) is tight for every $d \leq 56$, against $\sqrt{2} \cdot N^{3/4} \approx 51$.

Other pair constraints. Two further families of pair-level constraints are valid for Sidon sets: $\sum_{\{x < y, x+y=s\}} \Gamma(x, y) + f(s/2) \leq 1$ (distinct sums), and the Sherali–Adams inequalities $\Gamma(x, y) \leq f(x)$ and $\Gamma(x, y) \geq f(x) + f(y) - 1$. Adding them lowers the optimum by 0.019 at $N = 12$ and by 0.006 at $N = 100$.

7.5 Beyond pairs

The following results are preliminary; the sizes involved are small.

Lemma 7.8. *Consider any relaxation with triple variables $T(x, y, z)$, standing for the probability that $x, y, z \in A$, that contains the lifted budgets*

$$\sum_x T(x, x + d, z) \leq f(z), \quad \sum_x [\Gamma(x, x + d) - T(x, x + d, z)] \leq 1 - f(z), \quad (7.11)$$

that is, (iv) multiplied by $\mathbb{1}_z$ and by $1 - \mathbb{1}_z$. If $\sum_x \Gamma(x, x + d) = 1$, then $\sum_x T(x, x + d, z) = f(z)$ for every z .

Proof. The two inequalities in (7.11) sum to (iv). ■

In words, wherever the difference budget is full, the number of pairs at difference d must be uncorrelated with every point indicator. In the extremal configuration every budget below H is full.

The independent atoms of (7.9) violate this: conditioning on an atom point adds its own pairs at short lags. Deterministic atoms have constant indicators, so they satisfy every constraint that involves them, at every level. They can be used at the pair level at $O(1)$ cost.

Deterministic atoms. Take for the two atoms the first and second halves, m points each, of the Sidon set of Ruzsa [16] for the smallest prime $p \geq 2m + 1$. Their difference sets D_1 and D_2 are disjoint, and $|D| = m(m - 1)$ for $D = D_1 \cup D_2$. The two halves sit at the two ends, the second translated rather than reflected, so that the pairs between them also have distinct differences.

At a lag in D the atoms already use the whole budget, so every other pair at that lag must vanish. Three changes to the construction (7.9) achieve this:

- *Hole.* Replace Φ_ℓ by the Toeplitz matrix with entries 1 at lags in $\{0\} \cup D$ and 0 elsewhere. Its symbol is at most $1 + 2|D|$, so the argument of Lemma 7.5 goes through with δ replaced by $c = 1/(1 + 2|D|)$.
- *Cap.* Cap the profile ψ at c . This makes $\tilde{\psi} = \psi$, so pairs of bulk points at lags in D vanish exactly.
- *Depleted zone.* Set $\psi = 0$ within $\text{span} + \max D$ of each end, so that no atom–bulk pair falls at a lag in D .

We checked the result in exact arithmetic where it matters: the atom–atom pair counts are integers, every one is at most 1, and at each lag carrying such a pair the other contributions to (iv) vanish to rounding (below 10^{-15}). Every other lag was checked after lowering δ by a relative 10^{-7} .

N	m	t	$\delta\sqrt{N}$	$\sum_x f(x)$	$\sqrt{N} + (2\sqrt{2}/3)N^{1/4} - \sum f$
10^4	7	1.2	1.0356	107.324	2.10
10^5	13	1.25	1.0358	330.962	2.03
10^6	22	1.414	1.0060	1027.411	2.40
10^7	40	1.5	1.0041	3213.196	2.10

Table 6. Feasible points of (7.3) whose atoms are genuine Sidon sets. The shortfall is $O(1)$, about 1.1 larger than in Table 4; its fluctuations come from the integer choice of m .

The extra cost has two sources. The depleted zones have length between $2.2\sqrt{N}$ and $2.7\sqrt{N}$. The cap removes the part of $\frac{1}{4}\mu_1$ above the bulk density, and a bulk density between 0.4% and 3.6% above $N^{(-1/2)}$ recovers it. The atoms of the extremal configuration can therefore be taken to be genuine Sidon sets, and any obstruction beyond the pair level must live in the bulk. At the pair level the bulk must use every difference below H once in expectation, except the set D of about $\sqrt{N}/2$ differences left to the atoms.

Three points on the interval. The three-point relaxation $\hat{h}_3(N)$ adds the triple variables T and six sets of constraints:

- positive semidefiniteness of the moment matrices localized at $z \in A$ and at $z \notin A$, for every z ;
- the lifted budgets (7.11);
- $T(x, x + d, x + 2d) = 0$, since a Sidon set contains no three-term progression;

- the Sherali–Adams bounds $T \leq \Gamma$ and $T(x, y, z) \geq \Gamma(x, y) + \Gamma(x, z) - f(x)$ and their permutations;
- $\Gamma \leq f$.

It was solved with Clarabel, and with SCS [15] for $N \geq 32$; the two agree to four decimals at $N = 26$.

N	12	18	20	26	32	40
$h(N)$	5	6	6	7	7	8
$\hat{h}_2(N)$	5.1194	6.1340	6.4291	7.2237	7.9189	8.7405
$\hat{h}_3(N)$	5.0000	6.0149	6.3271	7.1370	7.8554	8.6916
$\hat{h}_2 - \hat{h}_3$	0.119	0.119	0.102	0.087	0.064	0.049

Table 7. Three-point information lowers the relaxation by an amount that shrinks with N , where a second-order improvement would require growth like $N^{1/4}$.

At $N = 20$, the lifted budgets and the localized positive semidefiniteness do most of the work, and they are synergistic: each alone recovers about a third of the reduction. Excluding three-term progressions adds a little.

A cyclic model of the bulk. To isolate the bulk, consider translation-invariant relaxations on $\mathbb{Z}_n$ with $n = q^2 - 1$. The Sidon condition is taken in $\mathbb{Z}_n$, and every difference in a symmetric forbidden set F of $q - 2$ nonzero residues is excluded. Since $q(q - 1) = n - 1 - (q - 2)$, the counting bound is exactly q , and the pair-level relaxation attains it. We compare three forbidden sets of the same size:

- the *subgroup*: the nonzero multiples of $q + 1$, which are exactly the residues Bose’s sets [3] miss;
- *short lags*: $\pm 1, \dots, \pm \lfloor (q - 2)/2 \rfloor$, together with $n/2$ when q is odd;
- *random lags*: the same number of random pairs $\pm d$, again with $n/2$ when q is odd. At $q = 7$ these are $\pm 2, \pm 23$ and 24 ; at $q = 8$ the draw gave $\pm 5, \pm 10$ and ± 15 .

The four-point relaxation uses the moment matrix indexed by the subsets of size at most 2, with nonnegativity, monotonicity, and exact zeros on every set of at most four elements that contains a repeated difference or a forbidden one. It is block-diagonalized by the Fourier transform on $\mathbb{Z}_n$, and it returns exactly 4 and 5 on the Singer moduli $n = 13$ and $n = 21$ [18].

$n = q^2 - 1$	forbidden set	pair	three-point	four-point	largest genuine set
24	subgroup	5	5	5.0000	5
24	short lags	5	5	4.2983	4
24	random lags	5	5	4.2983	4
48	subgroup	7	7	7.0000	7
48	short lags	7	7	6.9720	6
48	random lags	7	7	6.9720	6
63	subgroup	8	8	8.0000	8
63	short lags	8	8	8.0000	7
63	random lags	8	8	8.0000	7

Table 8. The cyclic model. The largest genuine sets were determined by exhaustive search. At $n = 63$ the four-point relaxation was solved with SCS [15]; the returned solutions are feasible to within 3×10^{-8} , so, since the counting bound caps every relaxation at q , the values 8 are exact up to residuals of that order. For the subgroup at $n = 63$ the value follows without computation, since a genuine set of size 8 exists.

A set of size q whose differences cover all residues except the forbidden ones exists only for the subgroup; for the other two shapes the largest genuine set has $q - 1$ elements. The pair and three-point relaxations nevertheless return exactly q for every shape.

The four-point relaxation contains the Sidon condition in its native form, $a + b = c + d$, and at $n = 24$ it sees the obstruction: it closes 70% of the gap between the counting bound and the truth. At $n = 48$ it closes 2.8%, and at $n = 63$ none of it. Where it cuts, it separates the subgroup from the other two shapes, not short lags from random ones. (For odd q the forbidden sets contain $n/2$ and for even q they do not, so the three sizes are not perfectly comparable.)

So the obstruction is arithmetic and global. It is invisible at the pair and three-point levels, and what the four-point level captures of it disappears by $n = 63$. This suggests that no fixed level of the moment hierarchy captures it as n grows, although three small sizes cannot settle the question.

7.6 Consequences

This answers, at the first natural level, the question raised at the start of this section. The level-one moment relaxation (7.3) is an extension of the smoothing method that keeps the variance term and the difference slack. Its dual problem is the dual of a semidefinite

program, of which smoothing certificates are feasible points: the kernel autocorrelations are the multipliers of (iv), and the Cauchy–Schwarz direction is the positive semidefinite multiplier of (i). At moderate N its optimal dual is the extremal smoothing certificate up to local corrections.

Its second-order constant is $2\sqrt{2}/3$: at most that by Proposition 7.3, and within 1.03 of it at $N = 10^4, 10^5, 10^6$ and 10^7 by Theorem 7.6. Conjecture 7.7 would give the full limit. Keeping the variance term therefore gains only $O(1)$ as long as the argument sees one- and two-point statistics, and the evidence of Section 7.5 suggests that three- and four-point statistics add little.

In the cyclic model, what separates the relaxations from genuine Sidon sets is a design-type question: whether a near-perfect difference set with a prescribed set of missing differences exists. An argument that goes below $2\sqrt{2}/3$ probably has to use arithmetic structure of that kind, rather than the statistics of a fixed number of points.

8. Discussion

Consequences for Erdős Problem 30. Theorem 1 closes the question of how far the vector-valued smoothing method can take the second-order constant: exactly to $2\sqrt{2}/3$, and no further. The certificate of Theorem 2 is within 5×10^{-7} of that limit, so further optimization inside the framework is pointless. The qualitative conclusion — that the method cannot reach the $o(N^{1/4})$ regime, let alone the N^ϵ of Erdős Problem 30 — follows already from any positive lower bound; what Theorem 1 adds is the exact location of the wall.

What an improvement would have to retain. The framework discards two quantities: the variance term in Cilleruelo’s identity for the number of representations [5], which is nonnegative and is thrown away when the energy is bounded; and the slack in the bound $d(h) \leq 1$ on the multiplicity of small differences. Any argument that goes below $2\sqrt{2}/3$ must keep at least one of them. Section 7 shows that keeping them at the level of pair statistics is not enough: the natural relaxation that does so has the same second-order constant, $2\sqrt{2}/3$.

Why a renewal measure? The covering condition asks that the boundary profiles, averaged against the kernel over every window, be at least 1. Its dual is a measure μ on shifts with $\mu * R \geq 4$, which is a renewal-type equation. The extremal certificate makes it an equality. The uniform law and the linear kernel are tied together — $2x$ is the density of the maximum of two independent uniform variables — but this is an observation, not an explanation. The identity of Lemma 4.3 — that the renewal measure’s own

autocorrelation, corrected by its cumulative mass and a linear term, is constant on its first window — is what makes the bound blind to the shape of the kernels. We would like a conceptual proof of it; ours is a calculation. Lemma 7.4 gives one reading of it: a set whose density near each end follows $\frac{1}{4}\mu_1$ has a flat pair function on the first window.

Open questions. (i) Is $\gamma^*_{\text{sym}} = 2\sqrt{2}/3$? (ii) Do other smoothing schemes — windows of varying width, weighted energies, higher-order Cauchy–Schwarz — have a similarly explicit limit? (iii) Can an argument that uses the arithmetic structure of dense Sidon sets, rather than the statistics of a fixed number of their points, go below $2\sqrt{2}/3$? Section 7 shows that the pair-level relaxation cannot, and that three- and four-point relaxations capture little of the obstruction that separates it from genuine Sidon sets.

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Appendix A. Verified certificates

Each row was rationalized, its boundary program re-solved for the rational kernels, its boundary vectors rounded and repaired by a uniform increment, and all covering inequalities checked over $\mathbb{Q}$. The constants were then evaluated exactly.

certificate	m	L	K	inequalities	a	b	γ
two-sided, optimized	32	4	2	258	1.3131380851	0.6770798880	0.9429206688
two-sided, optimized	64	4	2	514	1.3230863497	0.6718702582	0.9428374024

certificate	m	L	K	inequalities	a	b	γ
two-sided, optimized	128	4	2	1026	1.3281707583	0.6692681652	0.9428162103
two-sided, optimized	256	4	2	2050	1.3305653396	0.6680561258	0.9428108643
symmetric, optimized	512	4	2	2049	1.3286071522	0.6690407889	0.9428108915
linear kernel 2x	256	8	1	4098	1.3333282471	0.6666717951	0.9428108696
linear kernel 2x	512	8	1	8194	1.3333320618	0.6666679802	0.9428095208

Table A1. Verified certificates. Because the optimum is flat in some directions, a and b of the optimized rows are not individually converged; only γ is.

Code and data. The optimizer, the exact-rational verifier, the continuum evaluator used for Section 6.3 and the certificates in plain text are available from the author and will be deposited with the published version.

Appendix B. Proof of Proposition 5.2

Throughout, C_0 and κ are the constants of Lemma 4.1(b), $\rho(x) = 2x$, and ω is the profile of Theorem 5.1 at the end that sees ρ , so $\omega = \frac{1}{4}(\mu_1 \star \rho) = \frac{1}{2}\rho + \frac{1}{2}(r \star \rho)$.

Lemma B.1. (a) $\omega(t) = e^\wedge t - 1$ on $[0,1)$, and $\omega(t) = \int_{-} \{t-1\}^\wedge \{t\} (t-y) r(y) dy$ for $t \geq 1$. (b) $0 \leq r \leq e$, so $|g| \leq 2$. (c) ω is C^1 on $(0,1)$ and on every interval $(k, k+1)$, $k \geq 1$, with $\omega'(t) = e^\wedge t$ on $(0,1)$ and $\omega'(t) = r(t) - r(t-1)$ on $(1,\infty)$ away from the integers; in particular $|\omega'| \leq 4$. (d) For $t \geq 1$, $|\omega(t) - 1| \leq \frac{1}{2} \sup_{[t-1,t]} |g| \leq \frac{1}{2} C_0 e^\wedge \{-\kappa(t-1)\}$; in particular $0 \leq \omega \leq 2$ on $[1,\infty)$. (e) ω jumps by -1 at $t = 1$, from $e - 1$ to $e - 2$, and is continuous on $(1,\infty)$.

Proof. (a) For $t \geq 1$, $\rho(t) = 0$ and $\omega(t) = \int_0^1 x r(t-x) dx$; substitute $y = t-x$. (b) $r \geq 0$ as a renewal density. On $[0,1)$ $r(x) = e^\wedge x < e$, and on $[1,2)$ $r(x) = e^\wedge \{x-1\}(e-x) \leq e^\wedge \{e-2\} < e$. For $x \geq 1$, $r(x) = \int_{-} \{x-1\}^\wedge \{x\} r$ by Lemma 4.1(a), so r never exceeds its supremum over $[0,2)$. (c) Differentiating (a), $\omega'(t) = -r(t-1) + \int_{-} \{t-1\}^\wedge \{t\} r = r(t) - r(t-1)$, using Lemma 4.1(a) again; r is continuous except at 1, so this is continuous on each $(k, k+1)$. Then $|\omega'| \leq |g(t)| + |g(t-1)| \leq 4$ on $(1,\infty)$, and $\omega' = e^\wedge t < 4$ on $(0,1)$. (d) Since $\int_{-} \{t-1\}^\wedge \{t\} (t-y) \cdot 2 dy = 1$, $\omega(t) - 1 = \int_{-} \{t-1\}^\wedge \{t\} (t-y) g(y) dy$, and $\int_{-} \{t-1\}^\wedge \{t\} (t-y) dy = \frac{1}{2}$. (e) follows from (a) and (c): $\omega(1-) = e - 1$, and $\omega(1+) = \int_0^1 (1-y) e^\wedge y dy = e - 2$. ■

Construction. Fix $m \geq 2$, $L = \max(2, \lceil (2/\kappa) \log m \rceil)$ and $n = Lm$, and write $l_j = \lfloor j/m, (j+1)/m \rfloor$. The kernel is $p_i = \int_{-} \{l_i\} 2x dx = (2i+1)/m^2$ for $i < m$, so $a = m \sum_i p_i^2 = 4/3 - 1/(3m^2)$. At the end whose covering condition involves the reversed kernel, take $v_j = 1$ for every j . At the other end put $\bar{u}_j = m \int_{-} \{l_j\} \omega$ for $j < n$, $\bar{u}_j = 1$ for $j \geq n$, and

$$u_j = \bar{u}_j + \eta \ (j < n), \ u_j = 1 \ (j \geq n), \ \eta = 4/m^2 + \varepsilon_L, \ \varepsilon_L = \sup_{\{t \geq L-1\}} |\omega(t) - 1|. \quad (\text{B.1})$$

Covering. At the reversed end every covering sum equals $\sum_i p_{\{m-1-i\}} = 1$. At the other end write $C_q = \sum_i p_i u_{\{q+i\}}$ and $\bar{C}_q = \sum_i p_i \bar{u}_{\{q+i\}}$, and let $\pi_q = \sum_{\{i: q+i < n\}} p_i$ be the mass of the kernel inside the box at shift q . Then $C_q = \bar{C}_q + \eta \pi_q$, and it suffices to show $1 - \bar{C}_q \leq \eta \pi_q$ for $0 \leq q < n$; for $q = n$ every entry is 1.

Interior shifts, $0 \leq q \leq n - m$. Here $\pi_q = 1$ and the window $[q/m, q/m + 1]$ lies in $[0, L]$. Since $x \in I_i$ exactly when $q/m + x \in I_{\{q+i\}}$, and $\bar{p}_i = m p_i$ is the mean of ρ on I_i ,

$$\int_0^1 \rho(x) \omega(q/m + x) dx - \bar{C}_q = \sum_i \int_{I_i} (\rho(x) - \bar{p}_i)(\omega(q/m + x) - \bar{u}_{\{q+i\}}) dx,$$

because the cross terms integrate to zero on each cell. The left integral equals 1 by Theorem 5.1. On each cell $|\rho - \bar{p}_i| \leq 1/m$. The only discontinuity of ω , at $t = 1$, lies on a cell boundary, so Lemma B.1(c) gives $|\omega(q/m + x) - \bar{u}_{\{q+i\}}| \leq 4/m$. Summing over the m cells, $|1 - \bar{C}_q| \leq m \cdot (1/m) \cdot (1/m) \cdot (4/m) = 4/m^2 \leq \eta \pi_q$.

Far-end shifts, $n - m < q < n$. Every kernel cell with $q + i \geq n$ meets an entry equal to 1, so $1 - \bar{C}_q = \sum_{\{i: q+i < n\}} p_i (1 - \bar{u}_{\{q+i\}})$. Each cell $I_{\{q+i\}}$ with $q + i < n$ lies in $[L - 1, L]$, since $q/m > L - 1$, so $|1 - \bar{u}_{\{q+i\}}| \leq \varepsilon_L$ and $1 - \bar{C}_q \leq \pi_q \varepsilon_L \leq \eta \pi_q$. The point is that the deficit carries the factor π_q , which can be as small as $p_0 = 1/m^2$. Bounding it by ε_L alone would require $\eta \geq \varepsilon_L / \pi_q$, of order $m^2 \varepsilon_L$.

Cost. Only the second end contributes, and $b - 1 = (1/m) \sum_{\{j < n\}} ((\bar{u}_j + \eta)^2 - 1)$. By Cauchy-Schwarz on each cell $\bar{u}_j^2 \leq m \int_{I_j} \omega^2$, so $(1/m) \sum_{\{j < n\}} (\bar{u}_j^2 - 1) \leq \int_0^L (\omega^2 - 1) = -1/3 - \int_L^\infty (\omega^2 - 1)$, using $\int_0^\infty (\omega^2 - 1) = b - 1 = -1/3$ for the certificate of Theorem 5.1. For $t \geq L \geq 2$, Lemma B.1(d) gives $|\omega^2 - 1| \leq 3|\omega - 1| \leq (3C_0/2) e^{-\kappa(t-1)}$, so the tail is at most $(3C_0/(2\kappa)) e^{-\kappa(L-1)}$. The increment adds $(2\eta/m) \sum_{\{j < n\}} \bar{u}_j + \eta^2 L \leq 2\eta \int_0^L \omega + \eta^2 L \leq 2\eta(2L + 1) + \eta^2 L$, since $\omega \leq e - 1$ on $[0, 1]$ and $\omega \leq 2$ beyond. Hence

$$b \leq 2/3 + (3C_0/(2\kappa)) e^{-\kappa(L-1)} + 2\eta(2L + 1) + \eta^2 L. \quad (\text{B.2})$$

By the choice of L , $e^{-\kappa L} \leq m^{-2}$. By Lemma B.1(d), $\varepsilon_L \leq \frac{1}{2} C_0 e^{-\kappa(L-2)} \leq \frac{1}{2} C_0 e^{2\kappa} m^{-2}$, so $\eta \leq (4 + \frac{1}{2} C_0 e^{2\kappa}) m^{-2}$ and every term of (B.2) beyond $2/3$ is at most a constant times $(\log m)/m^2$. As $a \leq 4/3$, $a \cdot b \leq 8/9 + C(\log m)/m^2$. Finally $b \geq 8/(9a) > 0$ by Theorem 4.4, so Proposition 2.3 applies to these certificates, and together with Corollary 4.5 this gives $\gamma^* = 2\sqrt{2}/3$. ■

Remarks. The factor $\log m$ comes only from spreading η over the whole box. The interior deficits have a definite leading term, and a repair localized to the shifts where they occur gives $O(m^{-2})$, consistent with the behaviour $2/(9m^2)$ of the exactly optimal certificates in Table 2. Numerically, the construction of this appendix is feasible at $m = 32, 64$ and 128 . There its interior deficits are at most 1.8×10^{-5} at $m = 128$, against the bound $4/m^2 = 2.4 \times 10^{-4}$, and its far-end deficits relative to π_q are at most 1.1×10^{-7} , against $\varepsilon_L = 6.7 \times 10^{-7}$.

Appendix C. The smoothing inequality for two-sided certificates

We prove Proposition 2.3, following the proof of [10, Theorem 2.1] with the one change needed for asymmetric kernels, and make the $O(1)$ term explicit. In this appendix the kernels are indexed by $r = 1, \dots, K$, so that k can denote the size of a Sidon set, and after translation Sidon sets lie in $\{0, 1, \dots, N - 1\}$. For a two-sided certificate write C_r and D_r for the left-hand sides of the two systems in (2.3).

Lemma C.1. *Let $A \subset \{0, \dots, N - 1\}$ be a Sidon set with k elements, let a two-sided certificate at resolution (m, L, K) satisfy (2.3), and let $h \geq 1$ be an integer with $H = mh$ and $N \geq 2LH$. Then*

$$k^2 \leq (N - 1 + bH) \cdot (1 + (k - 1)a/H). \quad (C.1)$$

Proof. Kernels. Put $K_r(t) = p^{\wedge}(r)_i/h$ for $ih \leq t < (i + 1)h$, $0 \leq i < m$, and $K_r(t) = 0$ otherwise, so $K_r \geq 0$ and $\sum_t K_r(t) = 1$. Let $S_r(x) = \sum_{y \in A} K_r(x - y)$, which is supported in $J = \{0, \dots, N + H - 2\}$.

Energy. With $C_r(d) = \sum_t K_r(t)K_r(t + d) \geq 0$, we have $\|S_r\|^2 = \sum_{y, y' \in A} C_r(y - y') = kC_r(0) + \sum_{y \neq y'} C_r(y - y')$. Since A is Sidon, the differences $y - y'$ with $y \neq y'$ are distinct. As $C_r \geq 0$, the last sum is therefore at most $\sum_{d \neq 0} C_r(d) = 1 - C_r(0)$. Since $C_r(0) = \sum_i h(p^{\wedge}(r)_i/h)^2 = a_r/H$ with $a_r = m \sum_i (p^{\wedge}(r)_i)^2$, averaging gives

$$\sum_r \lambda_r \|S_r\|^2 \leq 1 + (k - 1)a/H. \quad (C.2)$$

Weights. Let $B_L = \{0, \dots, LH - 1\}$ and $B_R = \{N + H - 1 - LH, \dots, N + H - 2\}$, which are disjoint since $N \geq 2LH$. Define Q_r on J by $Q_r(x) = u^{\wedge}(r)_j$ when $jh \leq x < (j + 1)h$, and $Q_r(x) = v^{\wedge}(r)_j$ when $jh \leq N + H - 2 - x < (j + 1)h$ ($0 \leq j < Lm$ in both cases), and $Q_r(x) = 1$ otherwise. Then $\sum_x Q_r(x)^2 = N + H - 1 + h \sum_j [(u^{\wedge}(r)_j)^2 - 1] + h \sum_j [(v^{\wedge}(r)_j)^2 - 1]$, and averaging with $hm = H$ and the definition of b gives

$$\sum_r \lambda_r \|Q_r\|^2 = N - 1 + bH. \quad (C.3)$$

Covering. We claim that for every $x \in \{0, \dots, N - 1\}$

$$c(x) := \sum_r \lambda_r \sum_{t=0}^{H-1} K_r(t) Q_r(x + t) \geq 1. \quad (C.4)$$

The window $\{x, \dots, x + H - 1\}$ meets B_L only if $x < LH$, and meets B_R only if $x \geq N - LH$; since $N \geq 2LH$, it meets at most one of them. If it meets neither, every value of Q_r in it is 1 and $c(x) = 1$. If $x < LH$, write $x = qh + \tau$ with $0 \leq q < Lm$ and $0 \leq \tau < h$. For t in block i of K_r , the $h - \tau$ positions with $t - ih < h - \tau$ fall in block $q + i$ of Q_r and the other τ in block $q + i + 1$, where blocks with index at least Lm carry the value 1. Hence $c(x) = (1 - \tau/h)C_q + (\tau/h)C_{q+1} \geq 1$. If $x \geq N - LH$, write $x = N - 1 - (qh + \tau)$ with $0 \leq q < Lm$ and $0 \leq \tau < h$, and substitute $t' = H - 1 - t$. Then $N + H - 2 - (x + t) = qh + \tau + t'$, and t lies in block i exactly

when t' lies in block $m - 1 - i$, so $K_r(t) = p^\wedge(r)_{\{m-1-i'\}}/h$ when t' lies in block i' . The left-end computation, with $p^\wedge(r)$ replaced by its reversal and $u^\wedge(r)$ by $v^\wedge(r)$, gives $c(x) = (1 - \tau/h)D_q + (\tau/h)D_{\{q+1\}} \geq 1$. This is the only point at which [10] uses symmetry, through $K_s = K_{\{H-1-s\}}$.

Conclusion. Summing (C.4) over $x \in A$ and exchanging the sums gives $k \leq \sum_r \lambda_r \langle Q_r, S_r \rangle$. By Cauchy-Schwarz in the direct sum of K copies of $\ell^2(J)$, weighted by λ_r , $k^2 \leq (\sum_r \lambda_r \|Q_r\|^2)(\sum_r \lambda_r \|S_r\|^2)$, and (C.2) and (C.3) give (C.1). ■

Corollary C.2 (Proposition 2.3, explicit form). *Let a two-sided certificate satisfy (2.3) with $b > 0$, and put $\gamma = v(ab)$ and $N_1 = N + bm$. Then for all $N \geq N_0(a, b, m, L)$,*

$$h(N) \leq \sqrt{N_1} + \gamma N_1^{\{1/4\}} + \gamma^2/2 + \gamma^3/(8N_1^{\{1/4\}}) = \sqrt{N} + \gamma N^{\{1/4\}} + ab/2 + o(1). \quad (C.5)$$

Proof. Let $A \subset \{0, \dots, N - 1\}$ be a Sidon set with $k = h(N)$ elements. Weakening (C.1) gives $k^2 \leq N + bH + Nka/H + abk$. Take $h = \lceil H^*/m \rceil$ with $H^* = \sqrt{Nka/b}$. Then $bH < bH^* + bm$ and $Nka/H \leq bH^* = \gamma\sqrt{Nk}$, so $k^2 \leq N_1 + 2\gamma\sqrt{N_1 k} + \gamma^2 k = (\sqrt{N_1} + \gamma\sqrt{k})^2$. Thus $k \leq \sqrt{N_1} + \gamma\sqrt{k}$, so $\sqrt{k} \leq (\gamma + \sqrt{\gamma^2 + 4\sqrt{N_1}})/2$ and $k \leq \sqrt{N_1} + (\gamma^2 + \gamma\sqrt{\gamma^2 + 4\sqrt{N_1}})/2$. Now use $\sqrt{\gamma^2 + 4\sqrt{N_1}} \leq 2N_1^{\{1/4\}} + \gamma^2/(4N_1^{\{1/4\}})$. The hypothesis $N \geq 2LH$ of Lemma C.1 holds for large N : from $k(k-1)/2 \leq N-1$ we get $k \leq \sqrt{2N} + 1$, so $H < H^* + m = O(N^{\{3/4\}})$. ■

Relation to [10]. For a symmetric certificate with $u = v = w$, (C.1) is [10, (12)], and (C.2), (C.3), (C.4) are [10, (5)], [10, (10)] and [10, (6)]. Apart from the right-boundary step, the only difference is in the last step. In place of the choice of H in [10], which uses the Erdős-Turán bound to control k , the inequality $k \leq \sqrt{N_1} + \gamma\sqrt{k}$ makes the argument self-contained and gives the explicit constant $ab/2$.

Mechanical check. We implemented the construction literally at integer positions, and checked (C.2), (C.3), (C.4) and (C.1). The covering (C.4) was checked at every position, not only at elements of A , together with its identification with the interpolated discrete sums. The test used six certificates (random asymmetric kernels with one and two kernels, the linear kernel, and the optimized two-sided certificate at $m = 32$), five Sidon sets (Ruzsa's sets with 30, 46 and 60 elements in intervals of length 769, 2014 and 3349, an Erdős-Turán set and a greedy random set) and $h = 1, 2, 3$. Every check passed, the interpolation identity to 4×10^{-16} . Swapping u and v , or using one vector at both ends, makes the covering fail: its minimum drops to 0.44 for the linear kernel.